

Customer Behaviour Analysis Report

Business Insights

1. The dataset contains **250,000 customer transaction records**, providing a comprehensive view of customer purchasing behaviour. The data includes product categories, payment methods, customer demographics, and purchase amounts.
2. The data quality is good, with no duplicate records. Missing values were identified mainly in the **Returns** column and handled during data cleaning.
3. Product categories contribute differently to overall sales, indicating that some categories are more popular and generate higher purchase activity than others.
4. Customer purchase behaviour varies across age groups and payment methods. The visualizations help identify spending patterns and customer preferences.
5. Customer churn analysis shows the proportion of customers who stayed versus those who left, helping identify opportunities to improve customer retention and engagement.

Recommendations

1. Focus marketing campaigns on the product categories that generate the highest sales and customer interest.
2. Implement customer loyalty and reward programs to reduce churn and encourage repeat purchases.
3. Improve promotional offers for low-performing product categories to increase customer engagement.

4. Provide personalized product recommendations based on customer purchase history and preferences.
5. Continuously monitor customer behaviour, sales trends, and payment preferences to make data-driven business decisions and improve overall customer satisfaction.

Executive Summary

This project focused on analysing customer purchasing behaviour using a real-world dataset. The dataset was cleaned and explored using Python libraries such as Pandas and Matplotlib in Google Colab. Basic statistical analysis, missing value detection, duplicate checking, and data visualization were performed to understand customer behaviour and sales patterns.

The analysis identified differences in product category performance, customer spending habits, payment methods, and customer churn. Visualizations including bar charts, pie charts, histograms, and scatter plots were used to present the findings clearly. Based on the analysis, business insights and recommendations were developed to help improve customer engagement, increase sales, and reduce customer churn.

Overall, this project strengthened practical skills in data cleaning, exploratory data analysis (EDA), data visualization, and business reporting using Python.